

Home Work 6
II B. Tech. (Common)
Mathematics-II (NMCI102)

1. Find the eigenvalues and a basis of eigenvectors of the following matrices:

$$(i) \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix} \quad (ii) \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix} \quad (iii) \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix} \quad (iv) \begin{bmatrix} 2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 2 & 0 & 3 & 0 \\ 1 & 4 & 2 & -6 \end{bmatrix}.$$

2. (i) If $X = \begin{bmatrix} 9 \\ 8 \\ -2 \end{bmatrix}$ is an eigenvector of the matrix $A = \begin{bmatrix} 8 & -4 & 2 \\ 4 & 0 & 2 \\ 0 & -2 & -4 \end{bmatrix}$, then find the corresponding eigenvalue. Also, find the other eigenvalues of A without computing the characteristic polynomial. Finally, find all eigenvalues of $A^2 + A + 2I$ and A^{-1} (if invertible).
(ii) The eigenvalues of a matrix are 2, 3, 13 and 7. Then find the determinant and trace of the matrix.
3. Find the eigenvalues of the following matrices and also for each eigenvalue determine its algebraic multiplicity (AM) and geometric multiplicity (GM).

$$(i) \begin{bmatrix} 2 & 0 & 0 \\ 4 & 2 & 0 \\ 6 & 0 & 2 \end{bmatrix} \quad (ii) \begin{bmatrix} 1 & -5 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (iii) \begin{bmatrix} 3 & 0 & 0 & 0 \\ -6 & 4 & 1 & 5 \\ 2 & 1 & 4 & -1 \\ 4 & 0 & 0 & -3 \end{bmatrix} \quad (iv) \begin{bmatrix} -1 & 0 & 12 & 0 \\ 0 & -1 & 0 & 12 \\ 0 & 0 & -1 & -4 \\ 0 & 0 & -4 & -1 \end{bmatrix}.$$

4. Using Cayley-Hamilton theorem, find (i) A^8 , if $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$, (ii) A^4 , if $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$.

5. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, show that $A^n = A^{n-2} + A^2 - I$, $n \geq 2$. Hence find A^{50} and A^{100} .

6. Find eigenvalues of the matrix $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ and verify Cayley-Hamilton theorem for this matrix. Find the inverse of the matrix A and also express $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$ as a linear polynomial in A .

7. Find the characteristic polynomial of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence compute A^{-1} . Also find the matrix B such that $B = A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$.